

Overview

This is the second QSET EPS board for the mission planned for launch in 2028–2029. It was designed by Jaron Cyna (23wbjx@queensu.ca). Feel free to reach out to me if there are any questions regarding the board.

HIGH LEVEL BOARD VIEW:
This board is meant to provide a few key properties to the board:
1: Maximum Power Point Tracking (MPPT) to track and maintain optimal charging speed from solar panels
2: 3V3 and 5V lines to provide power to ADCS, OBC, and (possibly) comms
3: Telemetry and fault detection which is processed by the onboard MCU and sent to OBC for interpretation over CAN
3.1: Telemetry includes many integrated elements of the different IC's, external MCU watchdog, and a current sensor on all lines.
3.2: Fault detection also has integrated elements from the ICs, but the primary detection comes from the E-Fuses integrated throughout the board

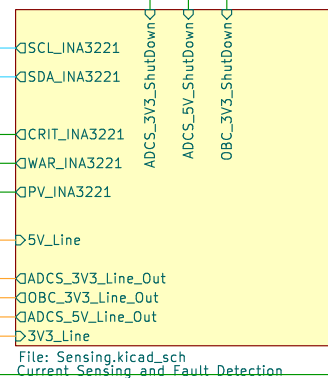
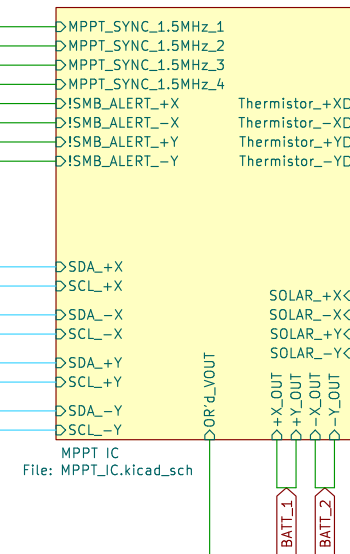
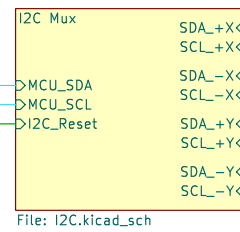
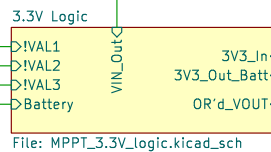
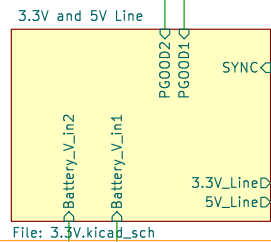
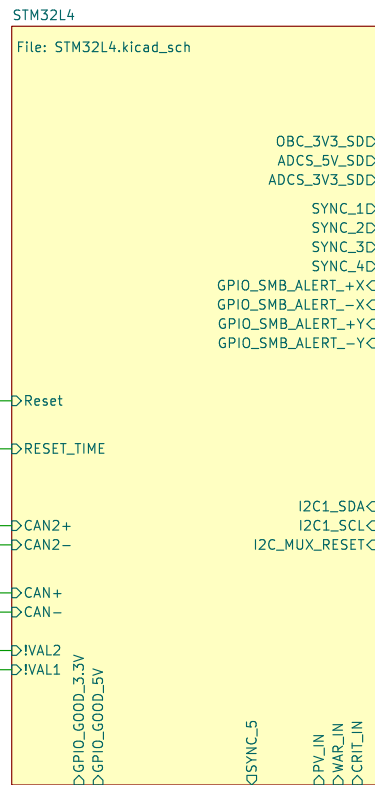
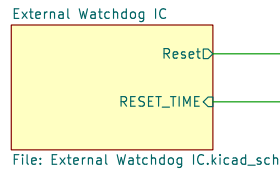
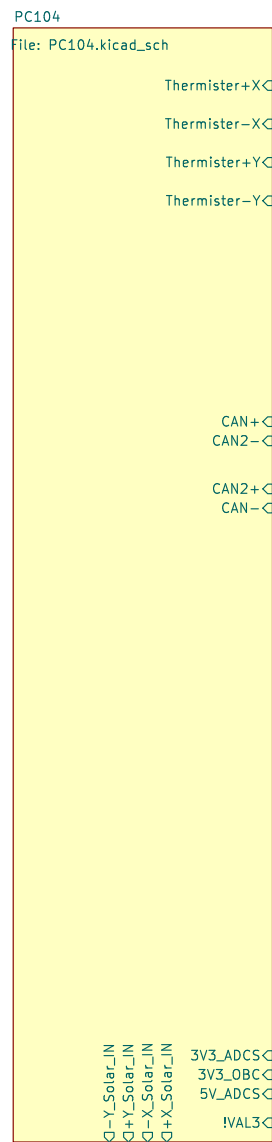
Connectivity

MCU

MPPT

Board Power

Safety



QSET (Jaron Cyna)

Sheet: /
File: EPS_mppt_board.kicad_sch

Title:

Size: A3
KiCad E.D.A. 9.0.0

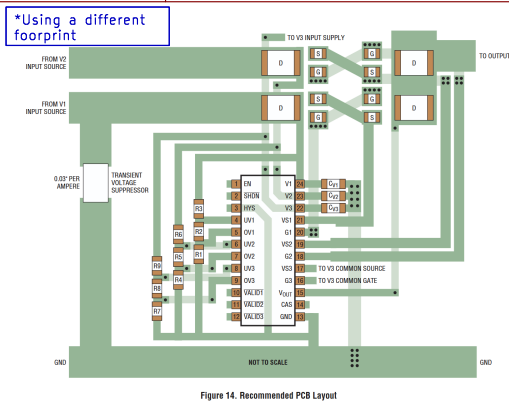
Date:

Rev:

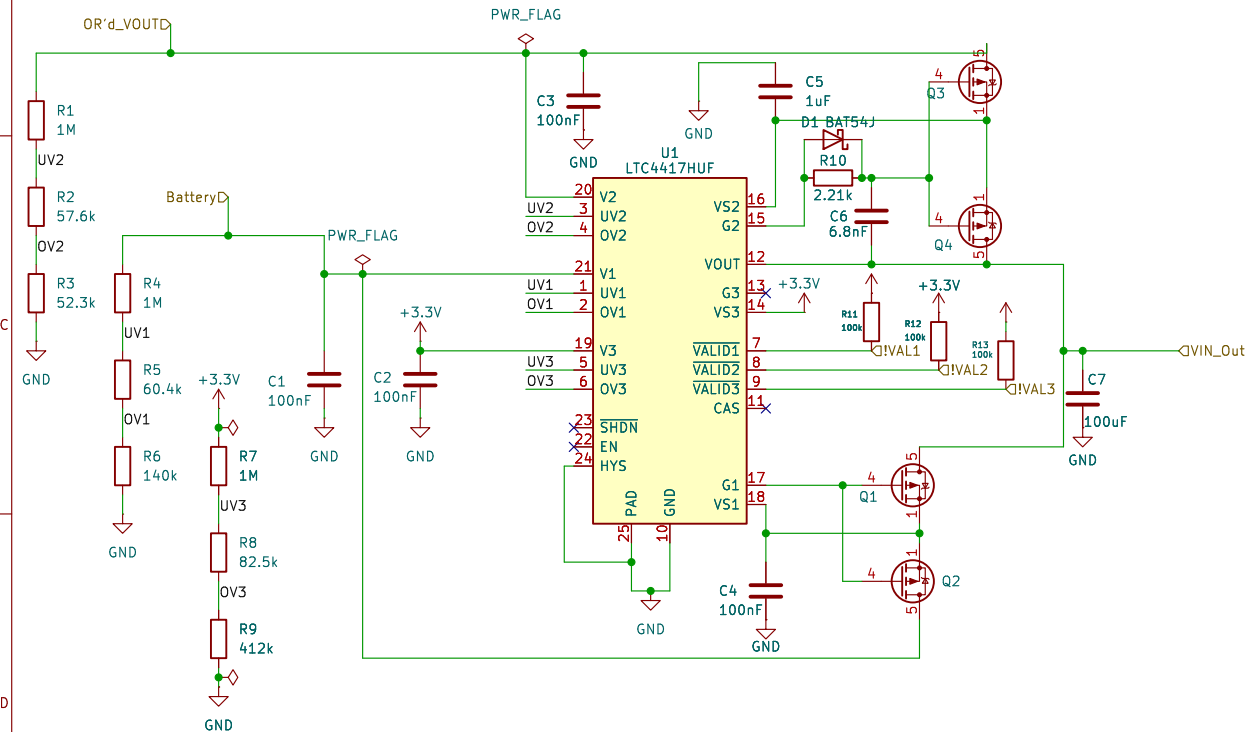
Id: 1/13

Ideal Diode ORing Circuit

*Using a different footprint



q current of =28 uA
which amounts to
=560 uW



1) Combined middle and bottom resistance:
 $R_{COMBINED} = R_{TOP} / (UV_{FALLING} - 1.00)$

2) Bottom grounding resistor:
 $R_{BOT} = (R_{COMBINED} + R_{TOP}) / OV_{RISING}$

3) Middle bridge resistor:
 $R_{MID} = R_{COMBINED} - R_{BOT}$

SOLAR LINE WINDOW CALCULATIONS (CHANNEL 1)
=====

Target Configuration:

- Low-Voltage Cutoff (UVLO Falling): 10.0V
- High-Voltage Cutoff (OVLO Rising): 21.0V
- LTC4417 internal threshold reference: 1.00V
- HYS Pin tied to Ground (Fixed 3% Hysteresis Mode)

BATTERY LINE WINDOW CALCULATIONS (CHANNEL 2)
=====

Target Configuration (2S Li-Ion Pack):

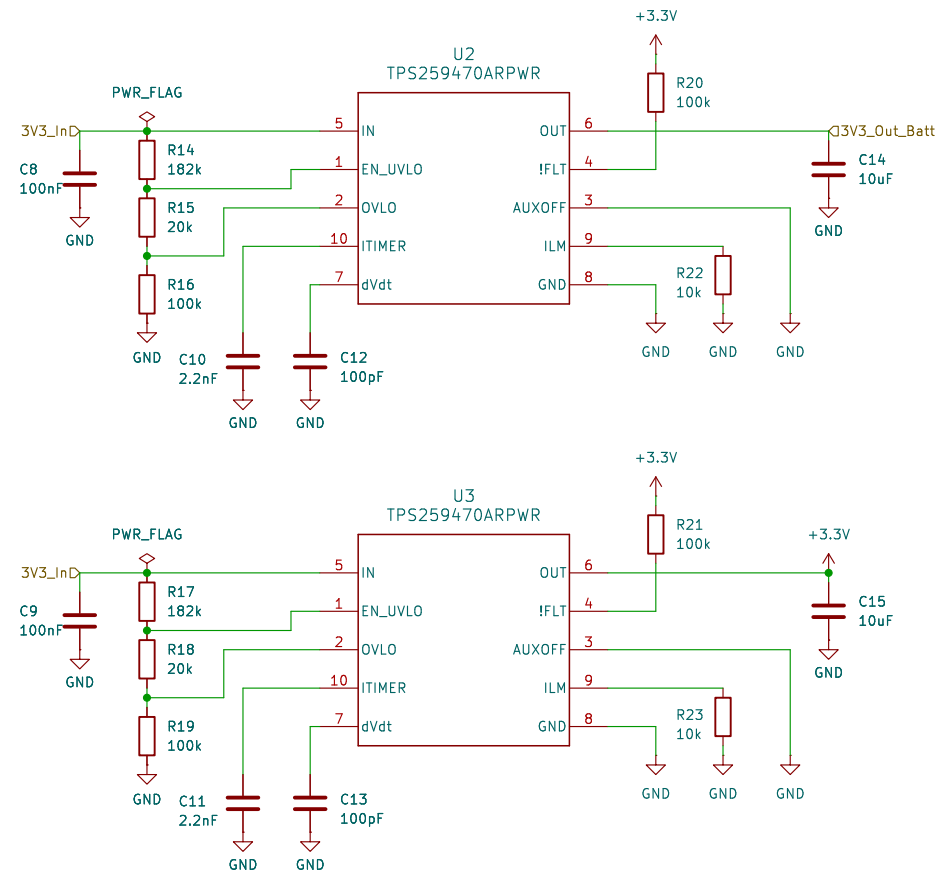
- Low-Voltage Cutoff (UVLO Falling): 6.0V
- High-Voltage Cutoff (OVLO Rising): 8.6V

+3.3V BUS SUPERVISOR WINDOW CALCULATIONS (CHANNEL 3)
=====

Target Configuration (Regulated Bus Health Check):

- Low-Voltage Threshold (UV Falling): 3.0V
- High-Voltage Threshold (OV Rising): 3.6V

E-Fuses



Current limit: $I_{lim} = 13.33/R_{ILM}(kohm) = 13.33/10 = 1.33 A$

Output Slew rate($dV/dt = 500/C_{dV/dt}(pF)$) = $500/100 = 5 V/ms$

for the Itimer:: $t = C_{ITIMER} * 0.2 = 2.2 * 0.2 = 0.44ms$

For OVLO and UVLO, this will turn on when the voltage is above 3.02V and below 3.63V

Changed from Dual LTC4412 Ideal Diode to single LTC4417 in Version 1.2
Added E-Fuse in Version 1.1
QSET (Jaron Cyna)
Sheet: /3.3V Logic/
File: MPPT_3.3V_logic.kicad_sch

Title: EPS Board 3.3V Power with Bootstrap

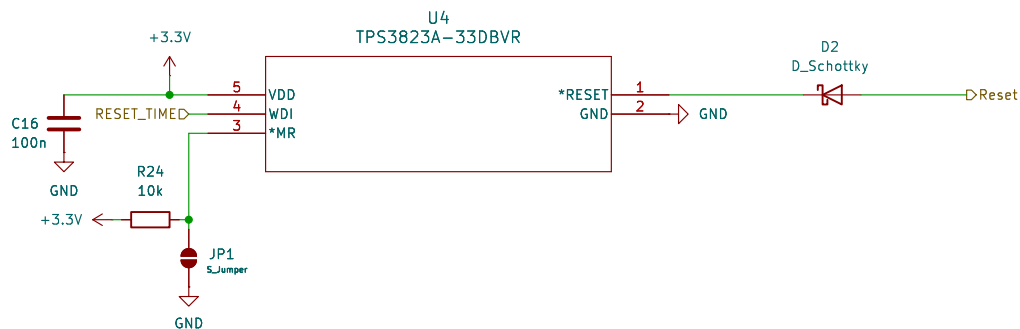
Size: A3 Date: 2026-04-28

Rev: 1.2

KiCad E.D.A. 9.0.0

Id: 1/13

Watchdog



QSET (Jaron Cyna)

Sheet: /External Watchdog IC/
File: External Watchdog IC.kicad_sch

Title: External Watchdog

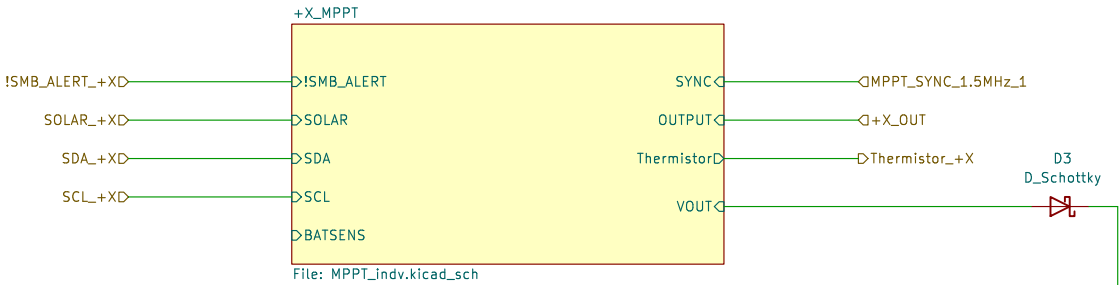
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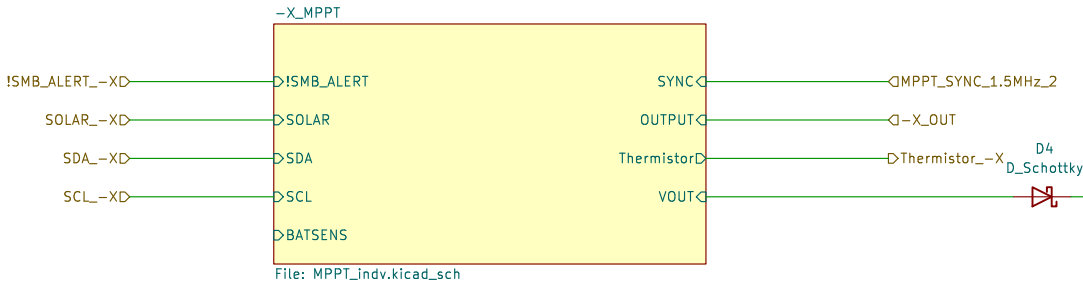
Rev: 1.0

Id: 2/13

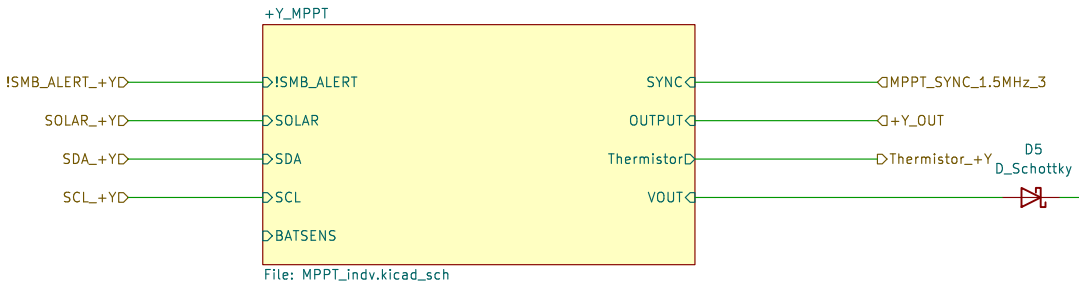
+X MPPT



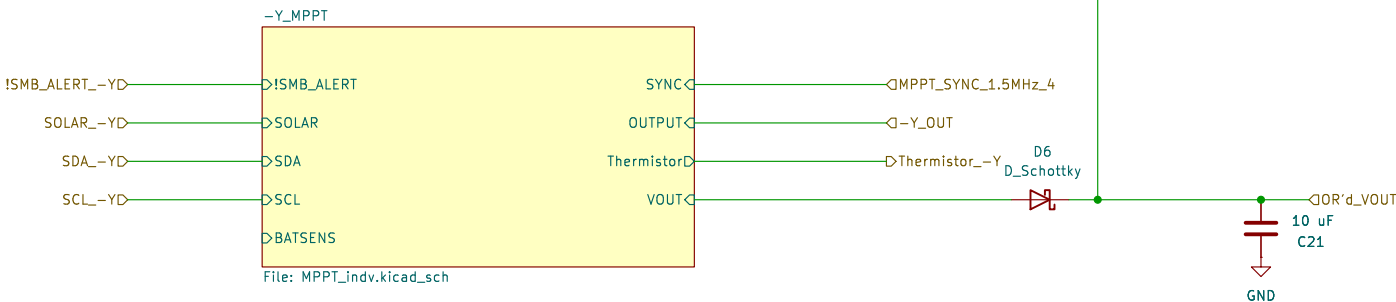
-X MPPT



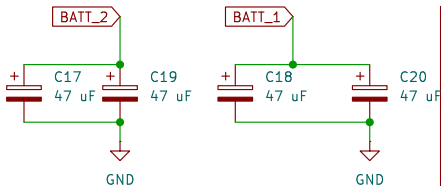
+Y MPPT



-Y MPPT



Notes:
-!SMB_ALERT will be tied to a External Interrupt Line (EXTI) on the MCU which will be triggered by a falling edge
- Think about ferrite Bead Placement on input
- Assumptions made when calculating inductor val:
15V solar input, 8.4V battery, 1.2A max charge limit
10uH (max I= 3A) (<30m ohms)



Switched to IC based MPPT from a discrete one for Power and Reliability

QSET (Jaron Cyna)

Sheet: /MPPT IC/

File: MPPT_IC.kicad_sch

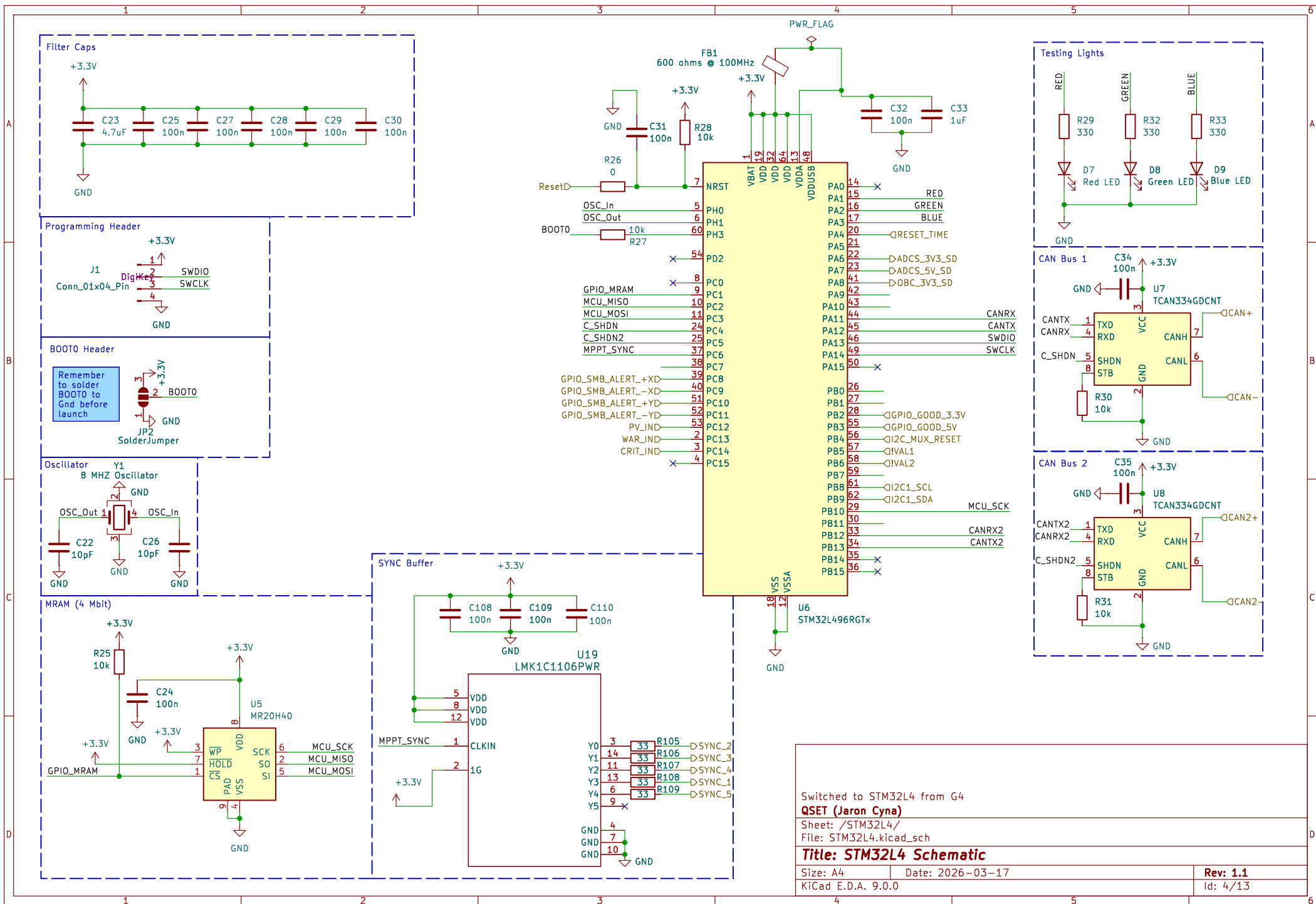
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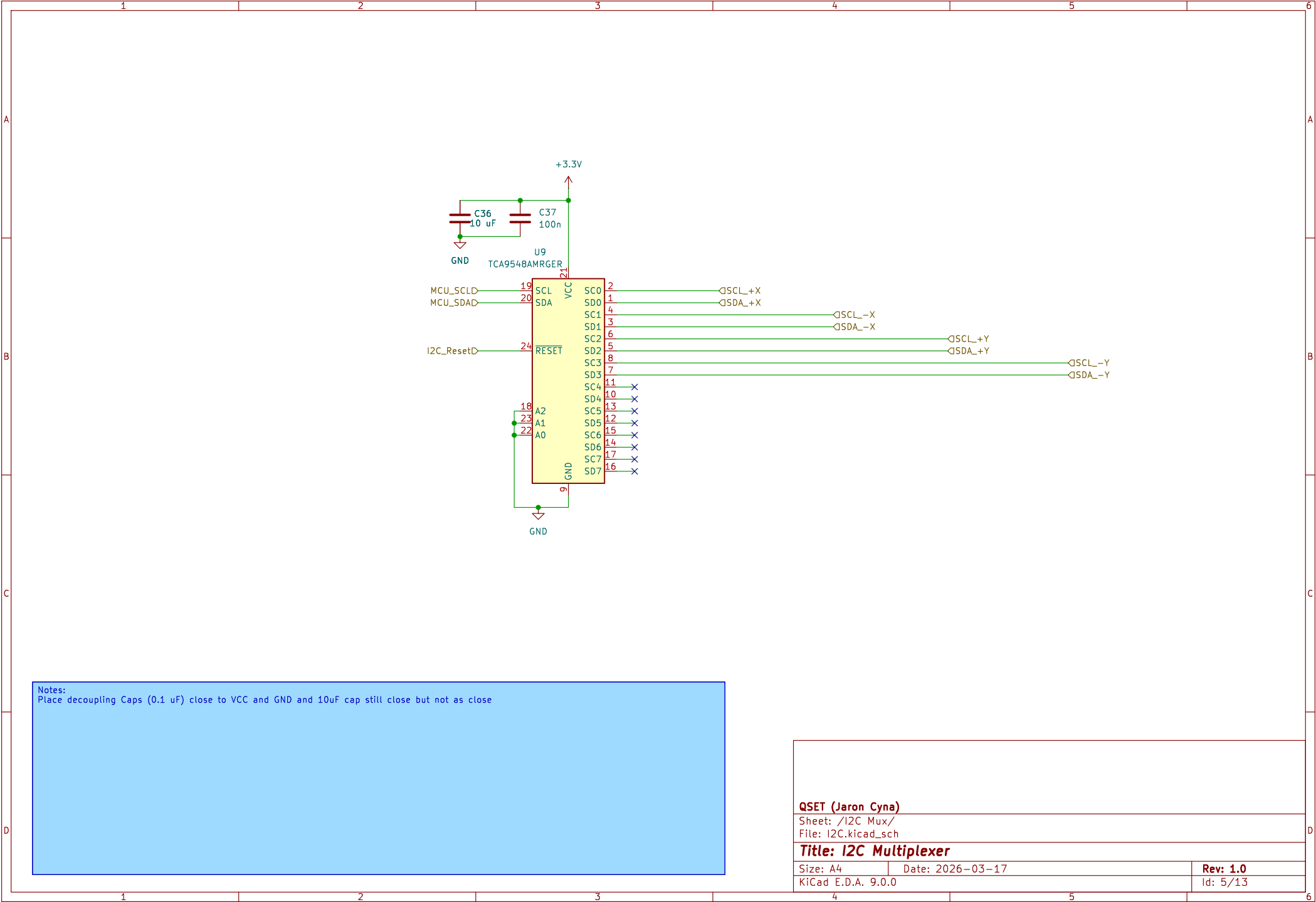
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Rev: 1.1

Id: 3/13

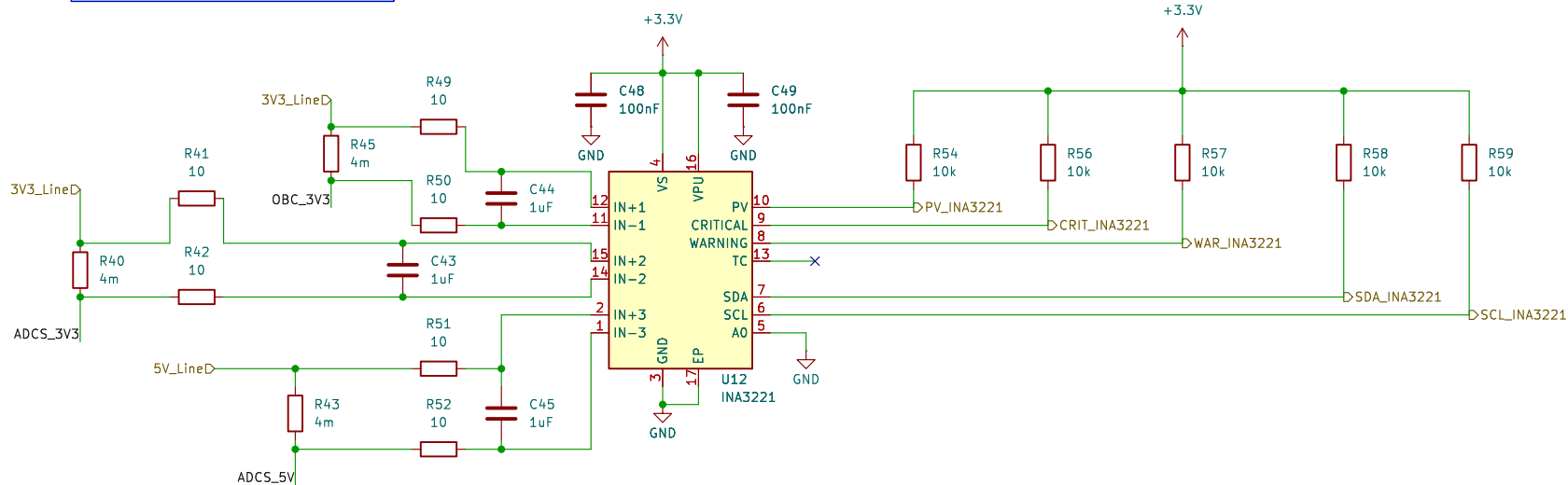




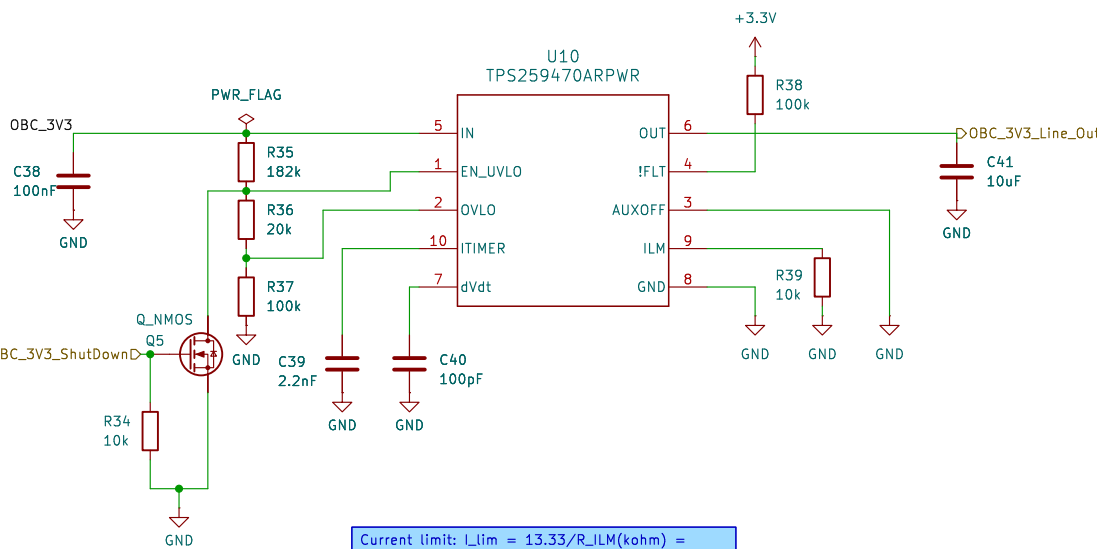
Current Sensing

By using 4m ohm shunts, the equation
 $I_{RES} = 40 \times 10^{-6} / R_{SHUNT} = 40 \times 10^{-6} / 0.004 = 0.01 \text{ A} = 10 \text{ mA}$
For the cutoff frequency of the RC filter, the equation is $f_c = 1 / (2 \times \pi \times R \times C) = 1 / (2 \times \pi \times 10 \times 10^{-6}) = 16 \text{ kHz}$

Use Kelvin Connections between 10 and 4m ohm resistors

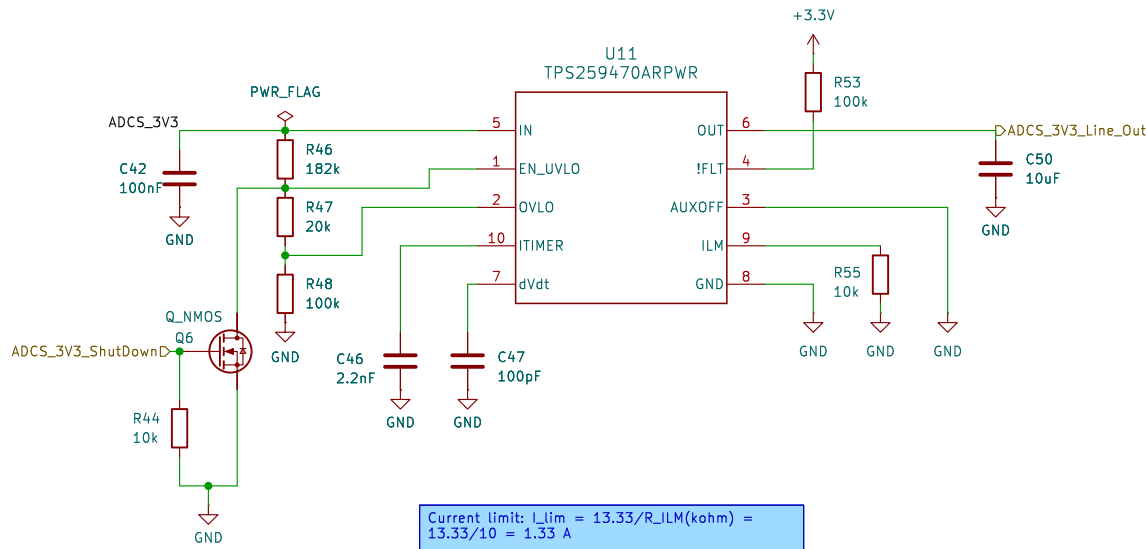


OBC 3.3V E-Fuse



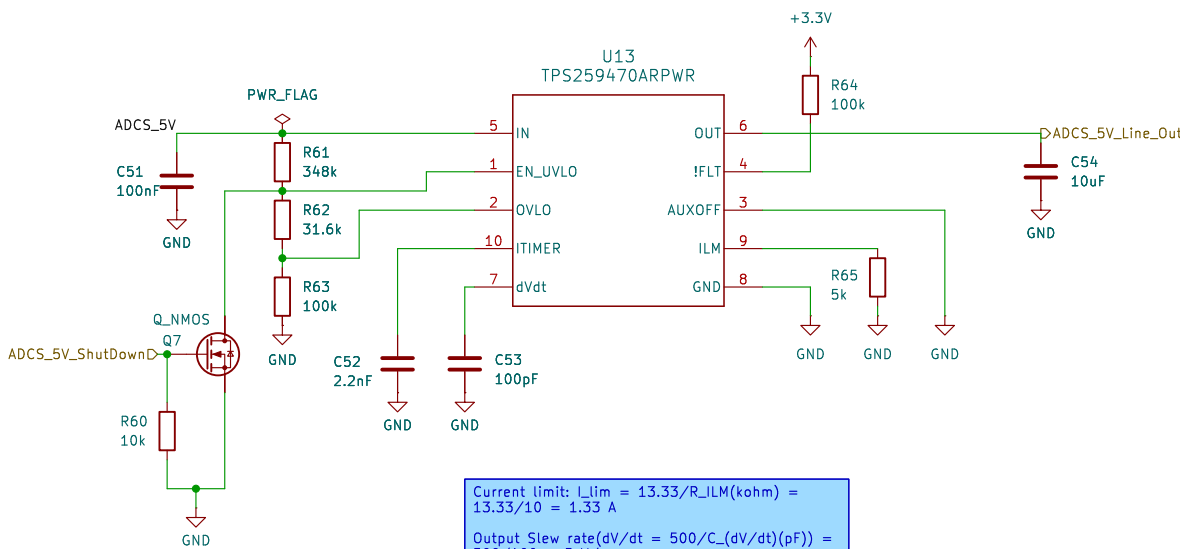
Current limit: $I_{lim} = 13.33 / R_{ILM}(\text{kohm}) = 13.33 / 10 = 1.33 \text{ A}$
Output Slew rate $(dV/dt = 500 / C_{(dV/dt)}(\text{pF})) = 500 / 100 = 5 \text{ V/ms}$
for the Itimer:: $t = C_{ITIMER} \times 0.2 = 2.2 \times 0.2 = 0.44 \text{ ms}$
For OVLO and UVLO, this will turn on when the voltage is above 3.02V and below 3.63V

ADCS 3.3V E-Fuse



Current limit: $I_{lim} = 13.33 / R_{ILM}(\text{kohm}) = 13.33 / 10 = 1.33 \text{ A}$
Output Slew rate $(dV/dt = 500 / C_{(dV/dt)}(\text{pF})) = 500 / 100 = 5 \text{ V/ms}$
for the Itimer:: $t = C_{ITIMER} \times 0.2 = 2.2 \times 0.2 = 0.44 \text{ ms}$
For OVLO and UVLO, this will turn on when the voltage is above 3.02V and below 3.63V

ADCS 5V E-Fuse



Current limit: $I_{lim} = 13.33 / R_{ILM}(\text{kohm}) = 13.33 / 10 = 1.33 \text{ A}$
Output Slew rate $(dV/dt = 500 / C_{(dV/dt)}(\text{pF})) = 500 / 100 = 5 \text{ V/ms}$
for the Itimer:: $t = C_{ITIMER} \times 0.2 = 2.2 \times 0.2 = 0.44 \text{ ms}$
For OVLO and UVLO, this will turn on when the voltage is above 4.41V and below 5.85V

At 1A per rail, hypothetically, the power loss of this circuit would be 32mW per line, which would be less than a percent of a 3.3V 1A or 5V 1A power.

QSET (Jaron Cyna)

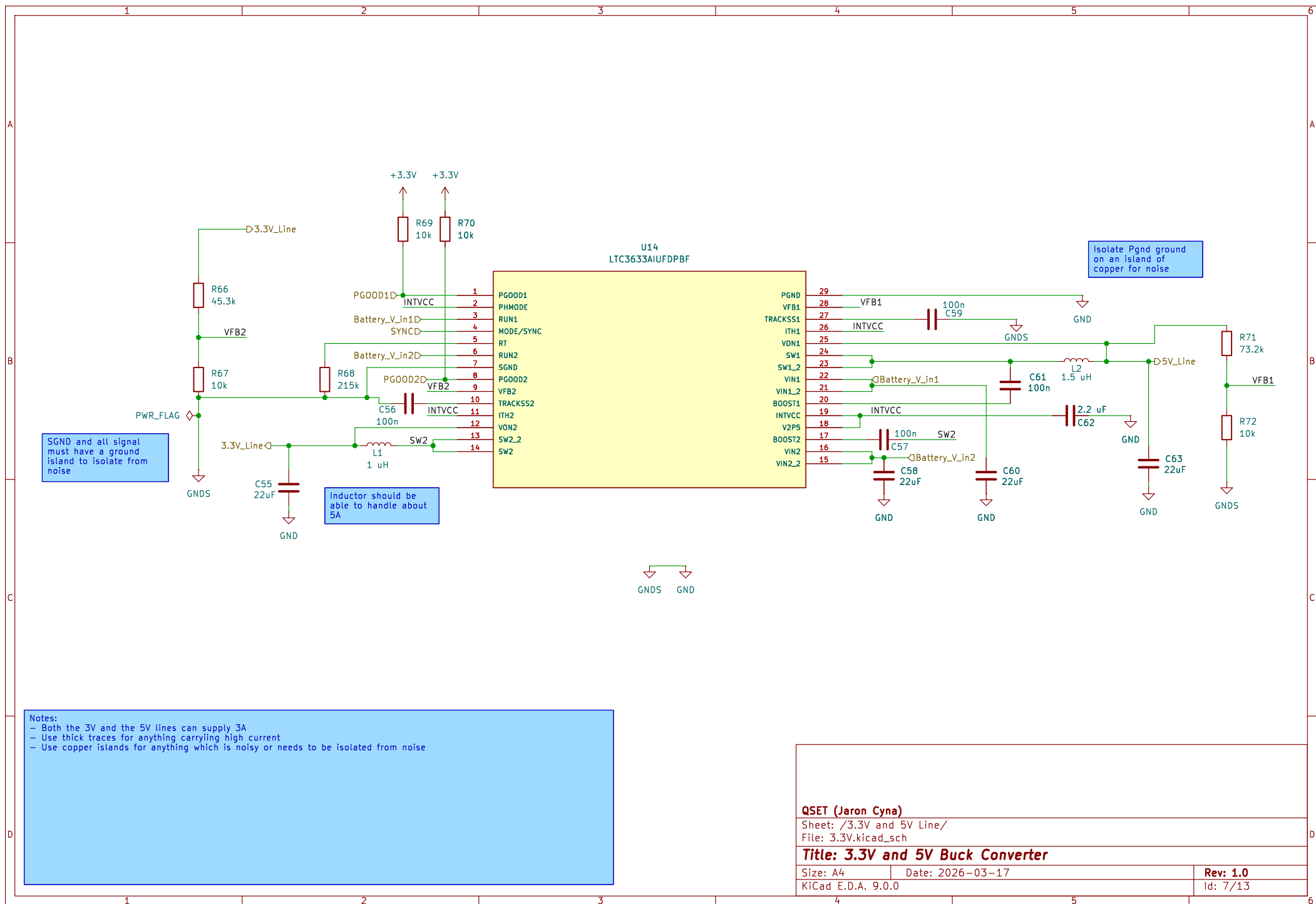
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File: Sensing.kicad_sch

Title: Current Sensing and E-Fuse

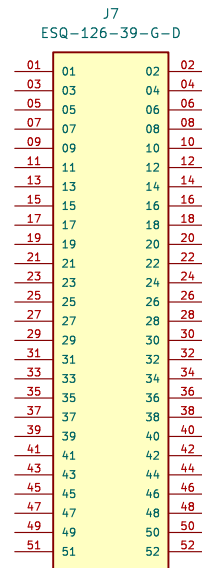
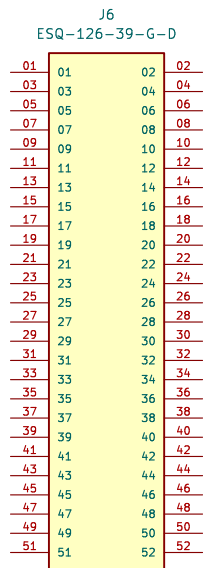
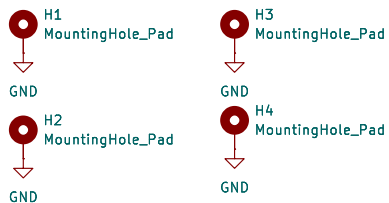
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Date: 2026-04-29

Rev: 1.0
Id: 6/13



NOT FINAL
PLACEMENT



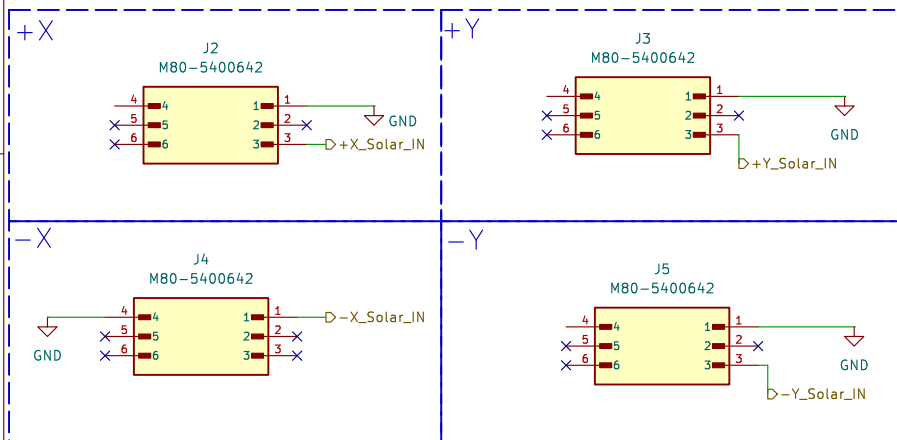
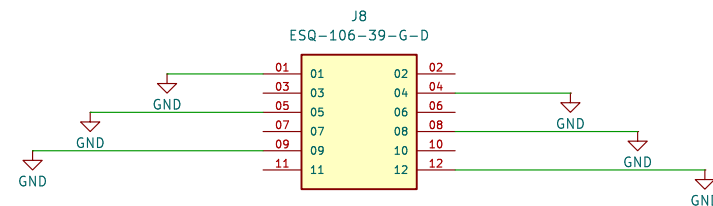
✕ CAN+
✕ CAN-

✕ CAN2+
✕ CAN2-

✕ IVAL3

✕ Thermister+X
✕ Thermister-X
✕ Thermister+Y
✕ Thermister-Y

✕ 3V3_OBC
✕ 3V3_ADCS
✕ 5V_ADCS



QSET (Jaron Cyna)

Sheet: /PC104/
File: PC104.kicad_sch

Title:

Size: A4

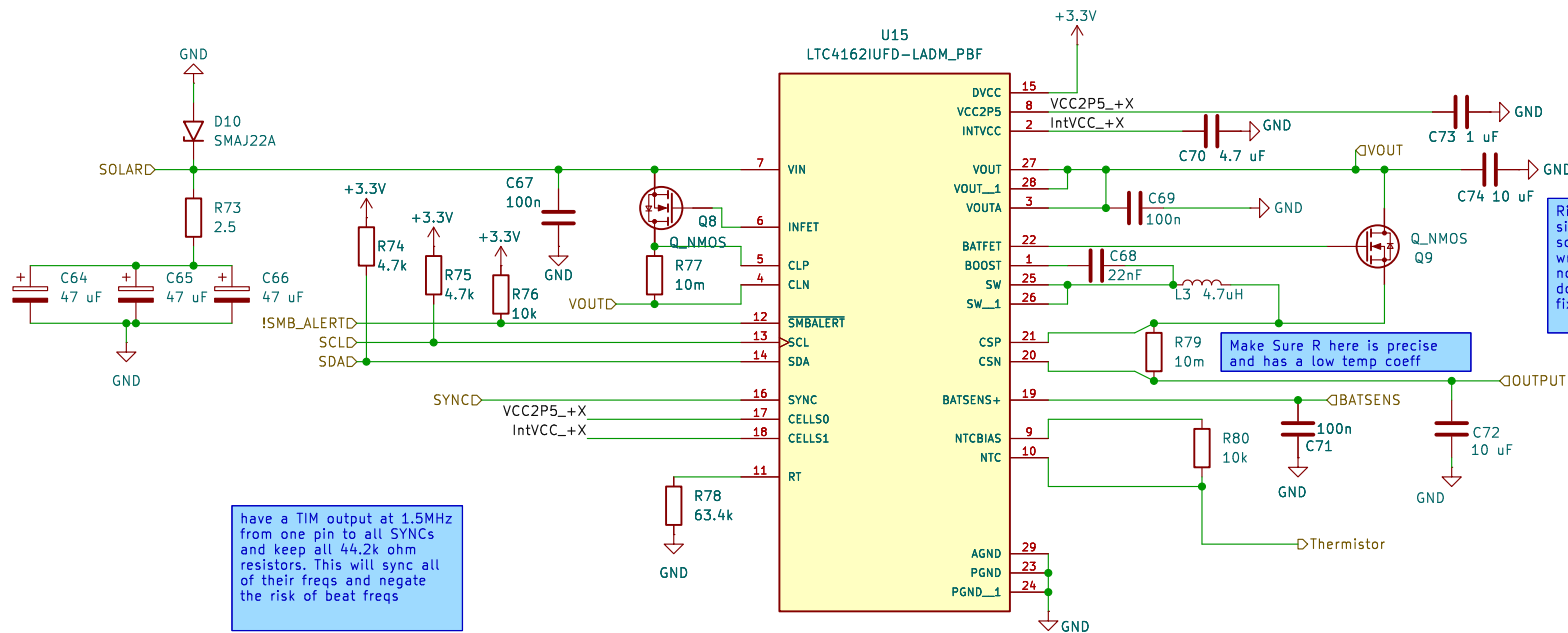
Date:

KiCad E.D.A. 9.0.0

Rev:

Id: 8/13

Maximum Power Point Tracker and Battery Charger



Some GND notes:
PGND (Pins 23,24): Power ground pins. These pins should be connected to a copper pour that forms the return for the V OUT bypass capacitor on the top layer of the printed circuit board.

AGND (Exposed PAD, Pin 29): Analog ground pin. This is the ground pin used to return all of the analog circuitry inside the LTC4162 and should be connected to an analog ground pour that is common with PGND (pins 23 and 24). It should also be connected to a ground plane on layer 2 of the PCB to which all of the analog components return such as the R T resistor and the INTVCC and VCC2P5 bypass capacitors.

Sheet: /MPPT IC/+X_MPPT/
File: MPPT_indv.kicad_sch

Title:

Size: A4

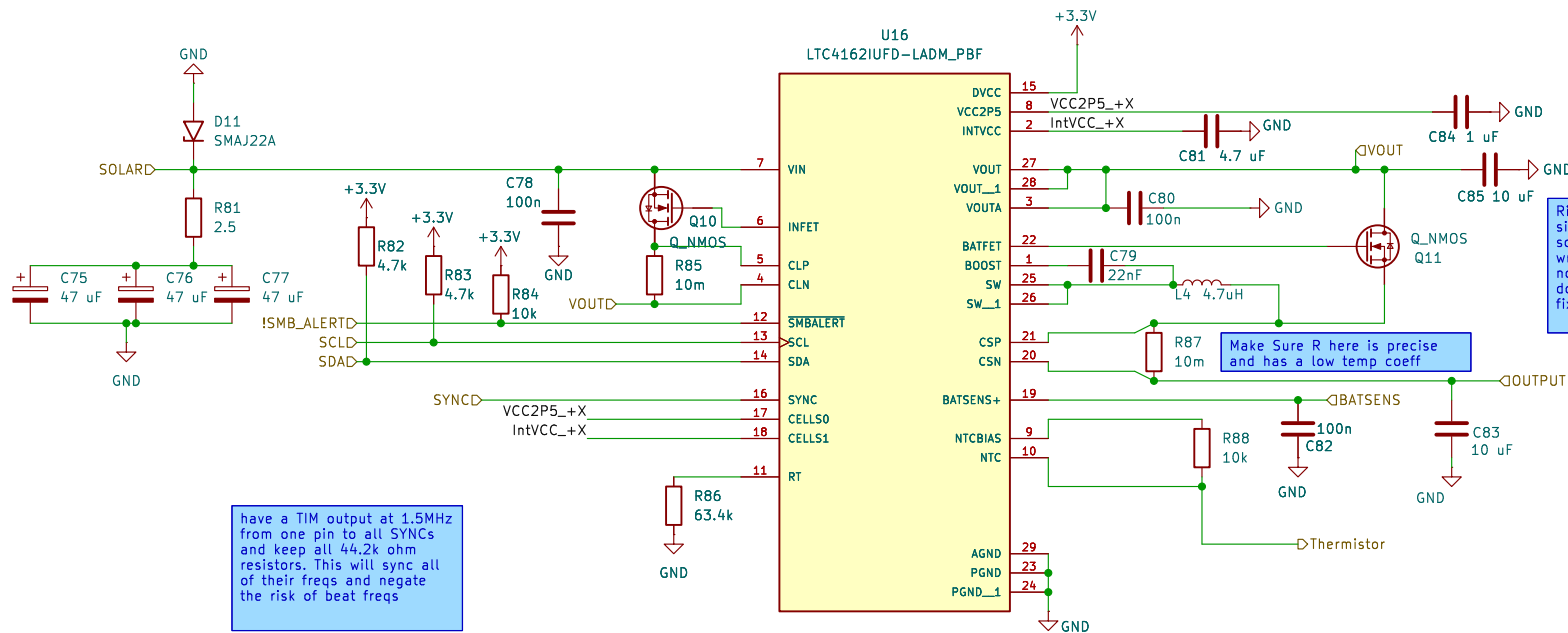
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KiCad E.D.A. 9.0.0

Rev:

Id: 9/13

Maximum Power Point Tracker and Battery Charger



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Sheet: /MPPT IC/+Y_MPPT/
File: MPPT_indv.kicad_sch

Title:

Size: A4

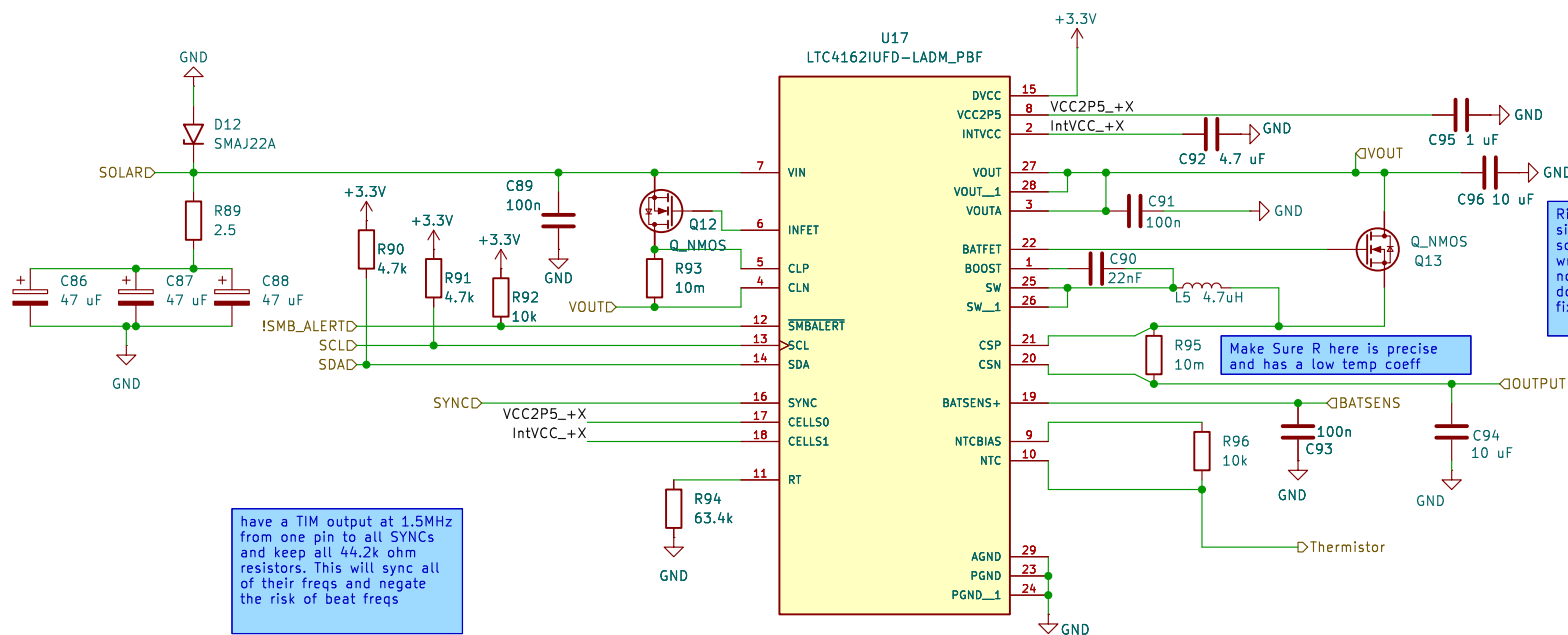
Date:

Rev:

KiCad E.D.A. 9.0.0

Id: 10/13

Maximum Power Point Tracker and Battery Charger



The NTC Pin Will likely be on the Battery Board for measuring battery Temp

Either through
PC/104 (Noisy) or
Molex Micro-Fit

If molex is used,
wire will need to be
twisted for noise,
and a cap will need
to be placed next
to ntc pin

have a TIM output at 1.5MHz
from one pin to all SYNCs
and keep all 44.2k ohm
resistors. This will sync all
of their freqs and negate
the risk of beat freqs

Some GND notes:
PGND (Pins 23,24): Power ground pins. These pins should be connected to a copper pour that forms the return for the V OUT bypass capacitor on the top layer of the printed circuit board.

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Sheet: /MPPT IC/-X_MPPT/
File: MPPT_indv.kicad_sch

Title:

Size: A4	Date:
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KiCad E.D.A. 9.0.0

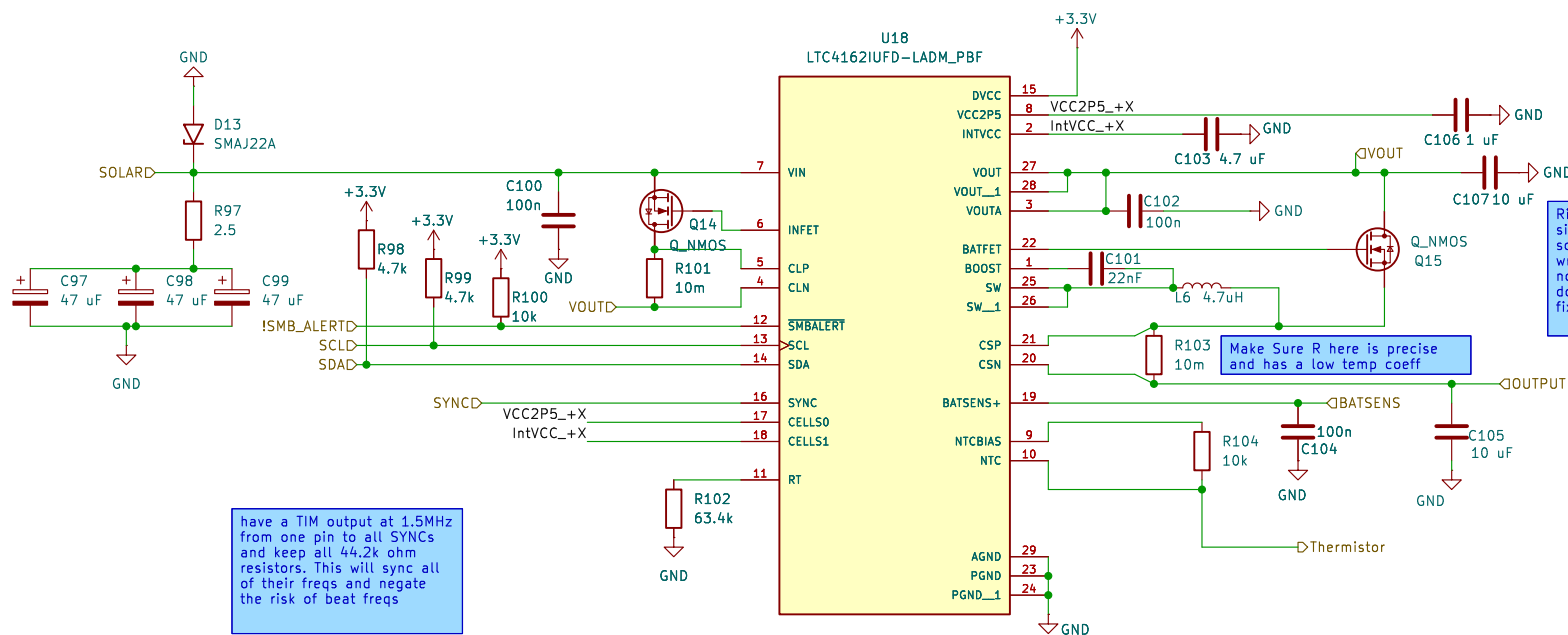
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Id: 11/13

Maximum Power Point Tracker and Battery Charger



The NTC Pin Will likely be on the Battery Board for measuring battery Temp

Either through
PC/104 (Noisy) or
Molex Micro-Fit

If molex is used,
wire will need to be
twisted for noise,
and a cap will need
to be placed next
to ntc pin

have a TIM output at 1.5MHz
from one pin to all SYNCs
and keep all 44.2k ohm
resistors. This will sync all
of their freqs and negate
the risk of beat freqs

Right now I have a signal for when something has gone wrong to the MCU but nothing the MCU can do about that, Need to fix that

Make Sure R here is precise and has a low temp coeff

Some GND notes:
PGND (Pins 23,24): Power ground pins. These pins should be connected to a copper pour that forms the return for the V OUT bypass capacitor on the top layer of the printed circuit board.

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Sheet: /MPPT IC/-Y_MPPT/
File: MPPT_indv.kicad_sch

Title:

Size: A4	Date:
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Id: 12/13